

B5 tier-assignment dossier: the consolidated case for full T5 within the pinned amended-class scope

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope — Purpose

Operator verdict #14 set B5 = T5-CANDIDATE and listed the full T5 assignment as an operator-review item. This dossier consolidates the decision inputs in one place. No tier change is performed here.

2. Pinned statement (the T5 object)

Statement (pinned scope). Within the matched second-cumulant B5 bookkeeping, for every pattern of the H-ADM-COH-amended admissible competitor class, at the anchor intensities $I \in \{4 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$ and the production operating point ($\mu^2 = 0.005$, A1-KERNEL-CONV convention), the beyond-layer off-diagonal correction obeys

$$|\delta F_{\text{off}}(\mathcal{P})| \leq \frac{K(n_{\text{pack}})(\lambda' I)^2 J_{\text{eff}}}{4(1 - a_0)} < m_{\text{STEP-5B}} = 0.00432,$$

with hardened margin floors $\times 59.4$ (anchor) and $\times 2.6$ (endpoint), band tops $\times 290.9$ / $\times 12.7$.

3. Scope fence and hypothesis accounting

- **Class:** H-ADM-COH-amended (operator-adopted, verdict #14); canonical quotient by the AddC energy-faithfulness ($c_{\text{ind}} = 30.1$; ratios $\times 898 / \times 139 / \times 33$) with the AddE depth-free pin ($c_{\text{total}} \leq 20$). Unrestricted class remains OPEN (DR-2, research-grade).
- **Order:** matched second cumulant; G2 bookkeeping closed at this level (Lemmas H/I/J; sextic fold $\varepsilon_4 \leq 0.16$). The third-cumulant landscape is quantified in the U4 assessment (anchor-feasible, endpoint-critical) and is OUTSIDE this statement.
- **Hypothesis fields:** the claim card carries NO named hypotheses — the conditionality lives in the CLASS DEFINITION (H-ADM-COH, adopted) and the ORDER fence (SC-SCOPE), both stated in the scope string, per the verdict-#14 bookkeeping. The 20/9 incidence route remains PROVISIONAL and UNUSED; no unpinned constant is in the closure path.
- **Constant provenance:** official $\sqrt{n}$ route with the operator-derived theorem-grade $c_R = 4\sqrt{14}$ (verdict #9/#10); H-KBAL structurally lifted ($64\sqrt{7}\sqrt{n}\log^2$ unconditional- amplitude theorem); packing $n_{\text{pack}} = 16/\theta_{\text{min}}^2 = 44/43/41$; $J_{\text{eff}} = 0.256/0.226$ (AddE refinement); $a_0 = 2\lambda' I / \hat{r}$ (P^2 structural floor).

4. Margin table (post-AddE, certified)

	$I = 4 \times 10^{-4}$	$I = 10^{-3}$	$I = 2 \times 10^{-3}$
closure margin (floor)	$\times 59.4$	$\times 8.8$ (AddD)	$\times 2.6$
criterion-band top	$\times 290.9$	—	$\times 12.7$
energy-faithfulness ratio	$\times 898$	$\times 139$	$\times 33$

5. Falsifier set (pre-registered, cumulative)

(i) An amended-class pattern inside a certified box whose exact off-diagonal Bloch correction exceeds the Lemma-B envelope; (ii) any pattern undercutting the $+0.00432$ band margin at the anchor; (iii) a sub-resolution restructuring shifting F by more than $c_{\text{ind}} I^2$ (refutes AddC); (iv) a resolution-respecting pattern with $n > n_{\text{pack}}$ (refutes the packing count); (v) a non-co-circular cluster pair with exact fiber multiplicity > 4 (refutes the AddE curvature pin); (vi) an amended-class transfer with $J(|t|) > J_{\text{eff}}$ (refutes the refinement); (vii) violation of the ℓ^1 identity $\lambda'(4S^2 - 2I)$ under splitting (Lemma C').

6. Reproduction contract

`python codes/vacuum/beyond_layer_gershgorin_bound.py` (v1.14.0): expected claims 192/192 PASS, exit 0, ~ 27 s; artefact [claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json](#) with the key fields: adversarial exact multiplicity 2; co-circular $K = 12.20 < 14$ (zero-slack absorption); $J_{\text{eff}} 0.256/0.226$; hardened margins $\times 59.4 / \times 2.6$.

7. Process record (verify-loop catches and audits)

Eight in-session verify-loop catches are documented across the chain (#1 ℓ^1 constant; #2 $c=0$ diagonal contamination; #3 banner-loss; #4 $c=0$ centre; #5 unproven $\sqrt{\nu}$ constant; #6 degenerate mirror config; #7 operator-caught $28/13 \rightarrow 20/9$ exponent; #8 draft assert contradicting Lemma C') plus 14 operator review verdicts — the audit density supporting reproducibility-grade confidence. The registered negative result (row/collar-ladder route fails $a < 1$ at production I) is on record in the chain.

8. TSv2 T5 artefact checklist

T5 requirement	Status
Pinned scope string on the card	PRESENT (section 3 fence = the card scope)
Closed within the pinned scope	YES per verdict #14 (gate row CLOSED-CONDITIONAL) + AddE polish (no unpinned constant)
Reproduction command + expected output	AVAILABLE (section 6; 192/192)
Falsifiers pre-registered	SEVEN (section 5)
Devil's-advocate records	in every chain note (main v1.x–v2.0, AddA–AddE)
Negative results registered	row-route refutation; H-GEN(2) falsification — both honest, on record

9. Numerical verification

No new numbers: every figure is quoted from the existing certified artefact (192/192) and the AddC/AddD/AddE notes. Quantitative sanity check (mandatory): cross-table consistency — the dossier margin table reproduces AddE’s hardened floors and AddD’s middle figure exactly; the falsifier set is the union of the falsification gates of the chain notes with no member dropped (checked one-to-one against the AddC, AddD, AddE footers and the card falsifier string); the checklist contains no item marked PRESENT without a disk pointer.

10. Devil’s-advocate (three objections; CLAUDE.md 6.3)

- (α) “*A dossier that consolidates only the favorable material is advocacy, not review input.*” DIS-MISSED by contents: the dossier includes the registered negative results (row route, H-GEN(2)), the thin endpoint ($\times 2.6$ floor flagged), the PROVISIONAL-and-unused status of the incidence route, the U4 endpoint-critical finding (which bounds what the T5 scope must NOT be read to imply), and the open unrestricted class.
- (β) “*The middle-intensity margin ($\times 8.8$) was not re-derived after AddE’s J_{eff} refinement — the table mixes AddD and AddE vintages.*” VALID-with-mitigation: correct and labelled in the table (‘(AddD)’); the AddE refinement only INCREASES the middle margin (smaller J_{eff} at larger minimum transfer), so quoting the AddD figure is the floor direction; the refreshed middle number is a one-line addition to the registered follow-up script.
- (γ) “*T5 assignment while the reproduction script was not re-run THIS session is premature.*” VALID-with-mitigation: the artefact cited is the canonical v1.14.0 result.json produced and operator-reviewed on 2026-06-05; nothing in the chain changed since; the dossier introduces no new numbers. The operator may of course condition the assignment on a fresh re-run (the contract in section 6 makes that a single command).

11. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / t5-assignment-dossier (unit U5)
Precise statement:	DOSSIER (no new mathematics): consolidated decision inputs for the T5-CANDIDATE -> T5 assignment -- pinned statement, scope fence, post-AddE margin table, seven pre-registered falsifiers, reproduction contract (192/192), process record (8 catches, 14 verdicts), TSv2 T5 artefact checklist (all PRESENT).
Scope:	Amended class; matched second cumulant; three anchor intensities; production operating point. Exactly the card scope string.
Dependencies:	The full B5 chain (main v1.0-v2.0 + AddA v1.0-1.3 + AddB-AddE) and verdicts #1-#14.
Evidence grade:	CONSOLIDATION of ANALYTIC + EXECUTED material.
Reproduction command:	python codes/vacuum/beyond_layer_gershgorin_bound.py

Expected output: claims 192/192 PASS (~27 s); key fields per section 6.

Falsification gate: The seven-item set of section 5 (cumulative union of the chain gates).

Tier before / after: B5 = T5-CANDIDATE before and after this note. PROPOSED: full T5 assignment -- OPERATOR SIGN-OFF REQUIRED.

No-overclaim statement: The dossier performs NO tier action and adds NO new result. T5 is scoped exactly to the pinned statement; it does NOT imply anything at third cumulant (U4), on the unrestricted class (DR-2), or off the anchor intensities. PDF build + linter PENDING operator-side.

Next required action: OPERATOR DECISION on the T5 assignment (optionally conditioned on a fresh 192/192 re-run). Mainline continues (U6+).